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Galactic panel
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Dynamical Atmosphere Evolution in Brown Dwarfs

Abstract

This research proposes a long-term near-infrared monitoring campaign of late-L and early-T brown dwarfs. The primary goal is to investigate variability in these objects at near-infrared wavelengths, which can provide insights into their atmospheric dynamics and evolution. By observing brown dwarfs over extended periods, we aim to uncover both rotationally modulated variability and potential global atmospheric changes. This study complements existing research that has focused on shorter timescales. By comparing light curves at different wavelengths, we can infer information about cloud layers, dust content, and atmospheric structure. Additionally, long-term trends can reveal global weather dynamics and the evolution of clouds over many rotation periods. This research will contribute to our understanding of the complex and dynamic atmospheres of brown dwarfs.

Observing Blocks

Instrument/Telescope	Req. time	Min. time	1^{st} Option	2^{nd} Option
ExTrA Spectrograph/ExTrA	10 nights	4 nights	Any Any	Any Any
CCD Speculoos/Speculoos	12 nights	4 nights	Any Any	Any Any

CoIs

Name	Institution	e-mail	Observer?
Nicola Astudillo-Defru	UCSC	nastudillo@ucsc.cl	False

Status of the project

- Past nights: 0
- Future nights: 0
- Long term: False

- Large program: False
- Thesis: False

List of Targets

ID	RA	DEC	Mag
2MASSJ0835425	08:35:42.53	-08:19:23.372	Jmag=13.2
36229BC	00:04:34.86	-40:44:06.463	Jmag=13.1
DENISJ025503	02:55:03.69	-47:00:51.356	Jmag=13.2
SIMPJ013656	01:36:56.56	+09:33:47.311	Jmag=13.3
Luhman16	10:49:14.02	-53:19:05.005	Jmag=11.5
2MASSJ0835425	08:35:42.53	-08:19:23.372	Jmag=13.2
36229BC	00:04:34.86	-40:44:06.463	Jmag=13.1
DENISJ025503	02:55:03.69	-47:00:51.356	Jmag=13.2
SIMPJ013656	01:36:56.56	+09:33:47.311	Jmag=13.3
Luhman16	10:49:14.02	-53:19:05.005	Jmag=11.5

SCIENTIFIC AIM AND RATIONALE

Brown dwarfs (BDs) are substellar objects. The high mass end of BDs is based on the minimum mass that would allow the fusion of protons to Helium in the object's core at around 75 M_J. The lowest mass BDs mark the border to the planetary mass regime at around 13 M_J(assuming the theoretical deuterium burning limit at this mass). Their similar chemical composition and size to large gaseous planets suggest the potential for cloud bands, storms, and other atmospheric phenomena. Photometric monitoring of BDs has revealed that these objects manifest a time-dependent variability, see e.g. Biller (2017) and Artigau (2018). One theory is that this long-term variability is due to global weather phenomena in the atmosphere. An inhomogeneous cloud deck, but stable over several rotation periods, causes a variation in flux, which is modulated through the BDs rotation (Marley et al., 2010). Another theory argues for non-uniform temperature profiles or perturbations in the atmosphere's temperature structure causing brightness fluctuations in BDs (Robinson & Marley 2014). Tremblin et al. (2016) argue that thermochemical instabilities cause the observed variability in these objects. Chemical abundance variations can form non-uniform surface opacities causing variabilities in the observation. The mechanism behind those variabilities is still under investigation. However, most of this studies focused on short-term variabilities

Initial studies in the optical wavelength range of detecting weather patterns through variabilities in ultra-cool dwarfs were focused on early to mid-L dwarfs (Martín & Zapatero Osorio 1997; Tinney & Tolley 1999; Bailer-Jones & Mundt 1999). Further investigations of early T dwarfs in the near-IR bands by Artigau et al. (2009) and Radigan et al. (2012) showed variabilities in these objects with periods between 2.4 and 7.7 hours. Later, Street et al. (2015) observed the Luhman-16 BD binary system photometrically in the Pan-STARRS-Z and SDSS-i' passbands with the LCO 1-m telescope network for 42 days. Luhman-16 components are of spectral types L7.5 and T0.5 (Burgasser et al. 2013). The authors detected a variability with an amplitude between 0.05 and 0.1 magnitudes in their light curves. The quasi-periodic modulation had periods ranging between 4.46 and 5.84 days, depending on the filter. They attribute this variability to the evolution of cloud features at varying latitudes on the surfaces. The TESS telescope obtained another light curve spanning several rotational periods of Luhman-16 (Apai et al. 2021) presented the light curve analysis. The authors found several peaks in the Lomb-Scargle periodogram, which they attributed to different causes, like the rotation of an inhomogeneous surface and planetary-wave beat pattern. Observing the ϵ Indi Ba and Bb brown dwarf binary (T1 and T6 components) with the DK154m telescope and the DEFOSC camera, Hitchcock et al. (2020) obtained I-band optical photometry. Based on the lack of variation and the existence of a long-term trend, they argue that the BDs in the ϵ Indi system are seen pole on, and the long-term trend is attributed to large-scale changes in cloud coverage. So far, most of the strongly variable dwarfs have been observed on short time scales (a few hours) in the near-IR. A typical rotational period of a BD is between 4 to 8 hours, and observations of a few hours span only one or two rotation periods. For long time-scale observations (a few weeks to a month), the number of observed dwarfs drops drastically to a few objects with most of them being observed over an extended period of time in the optical wavelength range missing the important additional information the near-IR wavelength range can provide.

Here, we propose long-term near-IR monitoriong of late-type L and early T dwarfs using the *ExTrA* and *SPECULOOS* facility with the aim to investigate variability at the near-IR wavelengths range. By monitoring a sample of brown dwarfs over extended periods, we can gain insights into both rotationally modulated variability and potential global atmospheric changes. This long-term perspective will provide valuable insights into the atmospheric dynamics and evolution of these fascinating objects. This approach complements existing studies that have focused on shorter timescales and specific spectral ranges. While JWST will provide unrivaled NIR spectra to understand the BDs' atmosphere better, our work will focus on the dynamical aspect of the atmosphere evolution over months and several rotation periods.

In order to study the atmosphere dynamics, we will use the time-series observations. We will use the light curves and estimate amplitudes, phases, periods, and any long-term trends. The *ExTrA* spectrograph will provide us with wavelength dependent light curves and *SPECULOOS* with single band light curves between 0.8-1.35 μm. We will conduct a Lomb-Scargle periodogram analysis and fit Gaussian Processes, e.g. Apai et al. (2021) and Vos et al. 2022. This allows us to model both the short and long-term modulations in the light curves. We can compare the amplitudes in different wavelengths from *ExTrA* which probe different atmosphere layers (Apai et al. 2013, Kellogg et al. 2017). The ratio of amplitudes in Ks and J can inform us about the dust in the atmosphere Artigau et al. (2009). Possible phase shifts in the periodic signal at different wavelengths can hint at multiple cloud layers (Biller et al. 2013). We can extract the amplitude of all light curves, compare them, and infer patterns in the 3D structure of brown dwarf atmospheres (Radigan et al. 2012). Long-term trends in the light curve might indicate global weather dynamics and inform about the evolution of clouds over many rotation periods. Also, slightly shifted peaks around the rotation period might indicate high-speed jets and zonal circulation in the BD's atmosphere (Apai et al. 2021). We will test the light curve for the presence of planetary wave patterns by fitting a sum of sine waves to our photometric time series as outlined by Apai et al. (2017).

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TECHNICAL DESCRIPTION

We propose to obtain time-series observations of 5 BDs with *ExTrA* in continuous blocks of 4-8 hours. A typical rotation period for these objects is between 6-8 hours. For each target we aim to obtain 4 of these continuous blocks separated by 1-2 weeks. Generally we will be able to observe ∼2 objects per night. Therefore, we request 10 nights with *ExTrA* for our program. The minimum amount of useful nights is 4, allowing us to obtain a complete series of observing blocks for ∼2 targets. We selected the brightest L/T dwarfs with $J \sim 13\,$mag. Observations can be done independent of moon phase.

We will observe a given field with all telescopes using the $4''$fibre with a spectral resolution of R∼20. *ExTrA* is equipped with 5 fibres, one will be centered on the target star and remaining 4 on comparisons stars in the same field. The comparison stars will track varying sky conditions and help us to identify possible systematics in the time-series observations. We use standard pipeline to extract flux and wavelength calibrate spectra.

For *SPECULOOS* observations we request time with the the 1028 × 1028 IR (InGaAs) camera SPIRIT using zYJ filter. We will schedule observations for ∼2 visible objects on 2 consecutive nights and a month later repeat the 2 nights' block for the same objects to obtain a longer baseline. To observe all 5 objects we estimate 12 nights, the minimum amount of 4 nights permits us to oberve ∼2 BDs. Exposure times of a single exposure in the series will be calculated for each object using the on-line exposure time calculator. Our faint brown dwarfs ($J \sim 13\,$mag) are outside the saturation limit for the shortest exposure time allowed. *SPECULOOS*'s field-of-view of $5.3' \times 6.6'$ will allow us to observe our target stars and a few reference stars, which will help us to remove some systematics affecting the photometry. Observations can be done independent of moon phase.

For the time-series analysis we will use standard Python packages.

CURRENT STATUS OF THE PROJECT AND STUDENT THESIS

This is a new project led by a Chilean institution.